

Perform 10 random draws. A possible sampling outcome could be: [elephant, pig, tiger, other, other, other, other, other, other, other]. The corresponding daily food estimate is:

$$\text{Sample Estimate} = \frac{50 + 20 + 10 + (7 \times 1)}{10} = 8.7 \text{ lb.}$$

This estimate is unbiased, meaning the expected value of the estimates equals the true average (8.7 lb). While there is variance across individual trials, the **standard deviation (std)** can be calculated as 4.7 lb for a 10-sample budget.

Increasing the sampling budget reduces variance. For example, with 20 samples, the **std** decreases to 3.4 lb. Meanwhile, Top-K with a budget of 20 adds 17 unique animals, yielding:

$$\text{TopK Average (K=20)} = \frac{(10 \times 50) + (10 \times 20) + (10 \times 10) + (17 \times 1)}{47} = 17 \text{ lb.}$$

Again, the Top-K estimate remains biased, significantly overestimating the average.

Note that this is intended as an intuitive example. For a detailed and formal derivation of the sampling methodology, please refer to [Kloek and Van Dijk \(1978\)](#); [Owen \(2013\)](#); [Lohr \(2021\)](#).

G Limitations and future work

MAGICPIG stores the offloaded KV cache and hash tables on CPU DRAM, which is unsuitable for serving scenarios with insufficient DRAM. KV cache quantization methods like QServe ([Lin et al., 2024](#)) and KIVI ([Liu et al., 2024b](#)) can help to reduce the KV cache memory. Currently, another limitation is that, we have not implemented MAGICPIG in prefilling stage, which is also an important direction in long context LLM serving. Applying more advanced LSH algorithms, such as Cross-polytope hash ([Andoni et al., 2015](#)), can reduce the size of hash tables while improving estimation accuracy. Building CPU-GPU pipelines ([He and Zhai, 2024](#)) and leveraging the new avx512_bf16 features of CPUs will improve efficiency. For higher-end GPUs with sufficient HBM, leveraging LSH to accelerate GPU attention computation is also an interesting topic, as GPU-friendly LSH algorithms and efficient GPU kernels ([Nguyen Mau and Inoguchi, 2020](#); [Pan et al., 2022](#)) are required to do sampling. Besides, how to automatically tune the LSH hyper-parameter (K, L) ([Lv et al., 2017](#)) is also an interesting future work.